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Magnetostratigraphy of the Lower Triassic beds from Chaohu (China) and its implications for the Induan–Olenekian stage boundary

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ABSTRACT

A magnetostratigraphic study was performed on the lower 44 m of the West Pingdingshan section near Chaohu city, (Anhui province, China) in order to provide a magnetic polarity scale for the early Triassic. Data from 295 paleomagnetic samples are integrated with a detailed biostratigraphy and lithostratigraphy. The tilt-corrected mean direction from the West Pingdingshan section, passes the reversal and fold tests. The overall mean direction after tilt correction is D=299.9°, I=18.3° (κ=305.2, α_{95}=1.9, N=19). The inferred paleolatitude of the sampling sites (31.6°N, 117.8°E) is about 9.4°, consistent with the stable South China block (SCB), though the declinations indicate some 101° counter-clockwise rotations with respect to the stable SCB since the Lower Triassic. Low-field anisotropy of magnetic susceptibility indicates evidence of weak strain. The lower part of the Yinkeng Formation is dominated by reversed polarity, with four normal polarity magnetozones (WP2n to WP5n), with evidence of some thinner (<0.5 m thick) normal magnetozones. The continuous magnetostratigraphy from the Yinkeng Formation, provides additional high-resolution details of the polarity pattern through the later parts of the Induan into the lowest Olenekian. The magnetostratigraphic and biostratigraphic data show that the conodont marker for the base of the Olenekian (first presence of *Neospathodus waageni*) is shortly prior to the base of normal magnetozone WP5n. This provides a secondary marker for mapping the base of the Olenekian into successions without conodonts. This section provides the only well-integrated study from a Tethyan section across this boundary, but problems remain in definitively relating this boundary into Boreal sections with magnetostratigraphy.

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1. Introduction

The global synchronous character of magnetic polarity reversals has made magnetostratigraphy a powerful tool for chronostratigraphic correlations between rocks from different environments, including those that are unfossiliferous or have poor fossil preservation. The Lower Triassic magnetostratigraphy has been studied in many places, where progress has been made in constructing a magnetic polarity scale for the Permian–Triassic boundary interval either from marine (Li and Wang, 1989; Steiner et al., 1989; Haag and Heller, 1991; Chen et al., 1994; Heller et al., 1995; Embleton et al., 1996; Heller et al., 1998; Zhu and Liu, 1999; Scholger et al., 2000; Gallet et al., 2000; Hounslow et al., 2008) or terrestrial rocks (Molostovsky, 1996; Szurlies et al., 2003; Steiner, 2006; Szurlies, 2007). In spite of this, discussions continue about how to precisely correlate marine and continental facies using magnetostratigraphy near the Permian–Triassic boundary (PTB) interval and the remainder of the Lower Triassic (Steiner, 2006; Szurlies, 2007; Hounslow et al., 2008). This in part reflects the debate about the placement of the base of the 2nd stage of the Triassic, the Olenekian, as well as differing quality, quantity and types of secondary biostratigraphic constraints in these early Triassic successions. In addition, paleomagnetic studies from some continuous marine sections have differing magnetostratigraphic records (e.g. the Induan global stratotype section and point (GSSP) at Meishan; Yin et al., 2001, 2005) perhaps because of undetected remagnetization (Li and Wang, 1989; Zhu and Liu, 1999; Steiner, 2006).

Investigations were carried out on the West Pingdingshan section to provide detailed biostratigraphic and lithostratigraphic information for the Induan–Olenekian boundary (Tong et al., 2003, 2005; Zhao et al., 2007, 2008), supported by carbon isotopic data (Tong et al., 2007). Based on this robust stratigraphic framework, a precise positioning of paleomagnetic data was achieved, resulting in a detailed magnetic polarity record. Along with biostratigraphic data, the magnetic record of these Lower Triassic sediments, is important in that they provide an integrated magnetostratigraphic and biostratigraphic reappraisal that allows the recognition of the Induan–Olenekian boundary in other sections. This is all the more important,

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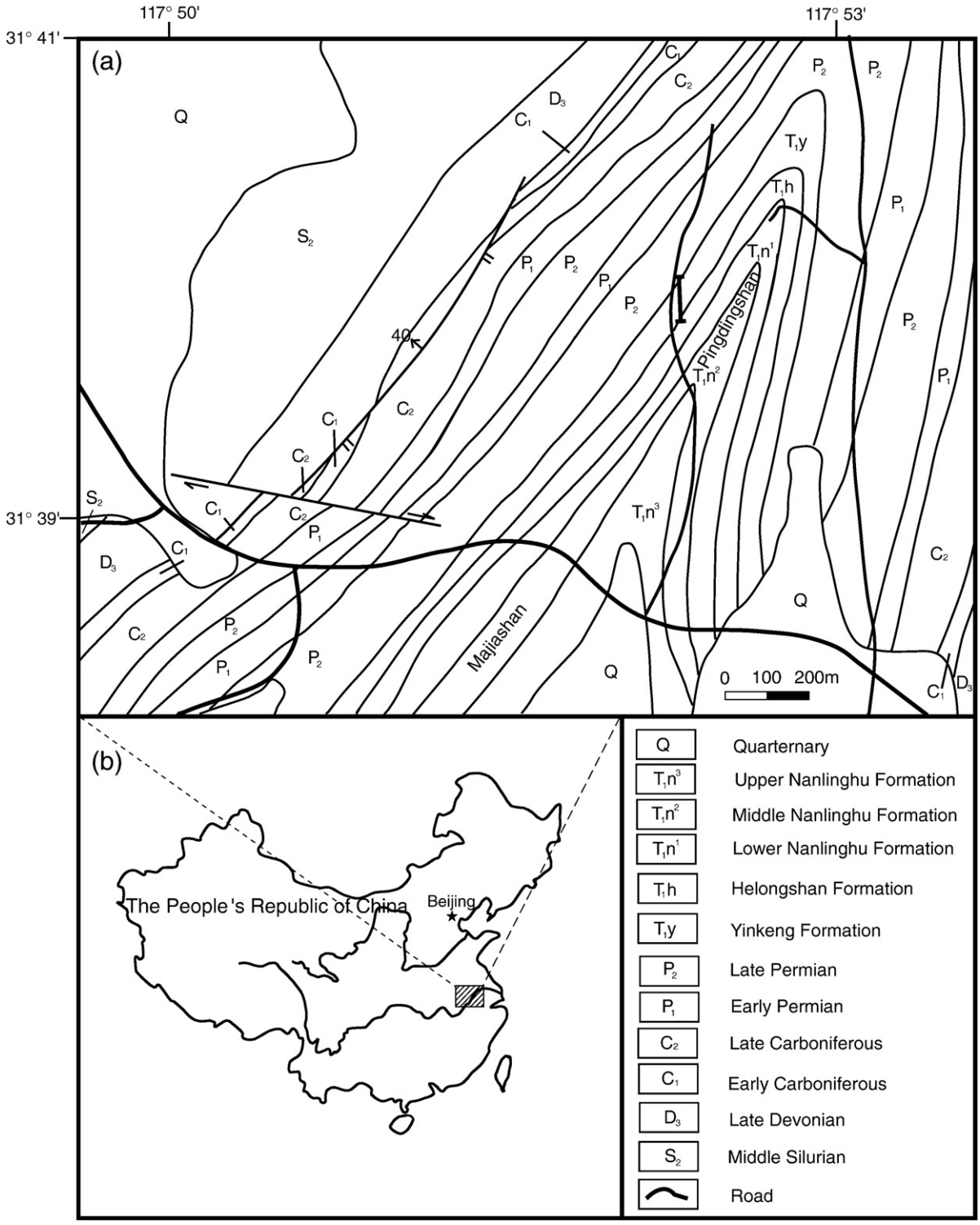


Fig. 1. (a) Simplified geological map of Chaohu area and (b) geographic location.

in that the ratified GSSP for the Olenekian at Mud (Spiti, India), is unlikely to ever have a magnetostratigraphy because of low grade metamorphism (Krystyn et al., 2007).

2. Geological setting and biostratigraphy

The West Pingdingshan section (location 31.6°N, 117.8°E), located near Chaohu city, Anhui province, China, consists of interbedded calcareous mudstones and limestones. The section is one of the well-exposed Lower Triassic successions in South China (Fig. 1), which was deposited in a carbonate ramp setting on the Lower Yangtze Block, within the low-latitude eastern Tethyan archipelago. The Lower Triassic sections near Chaohu city have been extensively investigated using a variety of detailed lithostratigraphic, biostratigraphic and chemostratigraphic tools (Tong et al., 2003, 2005, 2007; Zhao et al., 2007, 2008). The succession at Chaohu yields abundant fossils, which provide a comprehensive biostratigraphy of conodonts and ammonoids (Fig. 2).

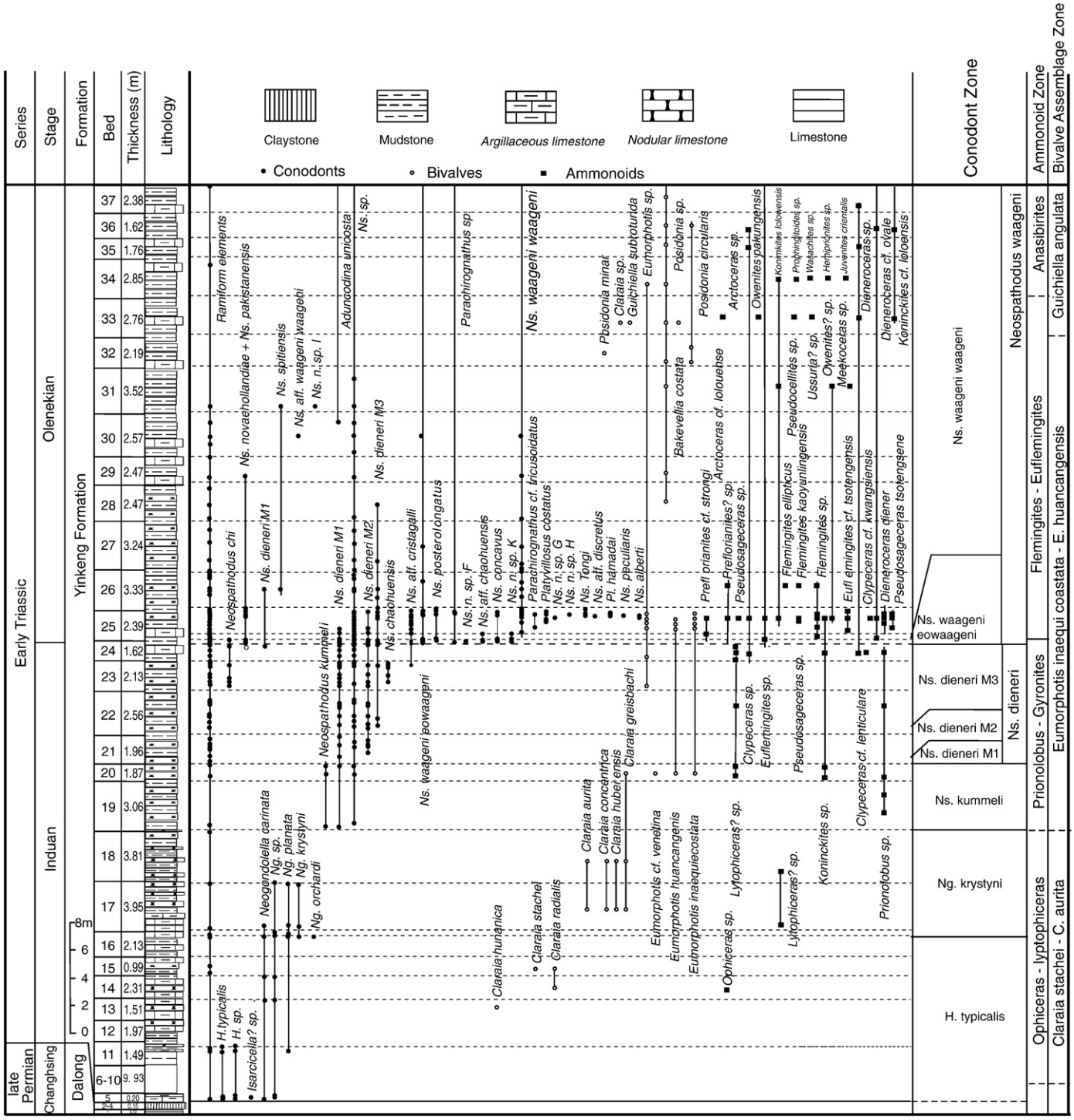


Fig. 2. Vertical range and zonation of conodonts, ammonoids and bivalves assemblage at West Pingdingshan section, Chaohu, Anhui Province. Modified after Zhao et al. (2008). Bold lines highlight the occurrence of the most important ammonoids and conodonts used to identify the Induan–Olenekian boundary.

Information is presented on a set of palaeomagnetic samples collected from the lower 44 m of the West Pingdingshan section, from the latest Permian into the Olenekian. As no conodont *Hindeodus parvus* has been found in the Chaohu sections, the exact P/T boundary is based on the "boundary sequence set" correlation (Peng et al., 2001). The latest Permian ammonoids occur in the highest beds of the Dalong Formation just 11.5 cm below the "boundary clay bed". Beds containing *Claraia* and *Ophiceras* occur about 50 cm above the "boundary clay bed", which is located at 0.33 m in the section. The I/O boundary in the WP Section, defined by the FAD of *Neospathodus waageni* Sweet (*sensu lato*), is at the base of Subbed 24–16 within a 20 cm bed of limestone. The Lower Triassic of Chaohu has received detailed investigation and the whole biostratigraphic succession has been established, including an 8-zone conodont succession, a 6-zone ammonoid succession and a 4-zone bivalve succession (Tong et al., 2005; Zhao et al., 2007). Conodont studies at Chaohu have resulted in discoveries of new forms, detailed understanding of basic successions and their global correlation (Zhao et al., 2007). As the primary marker defining the I/O boundary, conodonts in the WP Section and at the I/O boundary have been especially clarified. The *Neospathodus dieneri Zone*

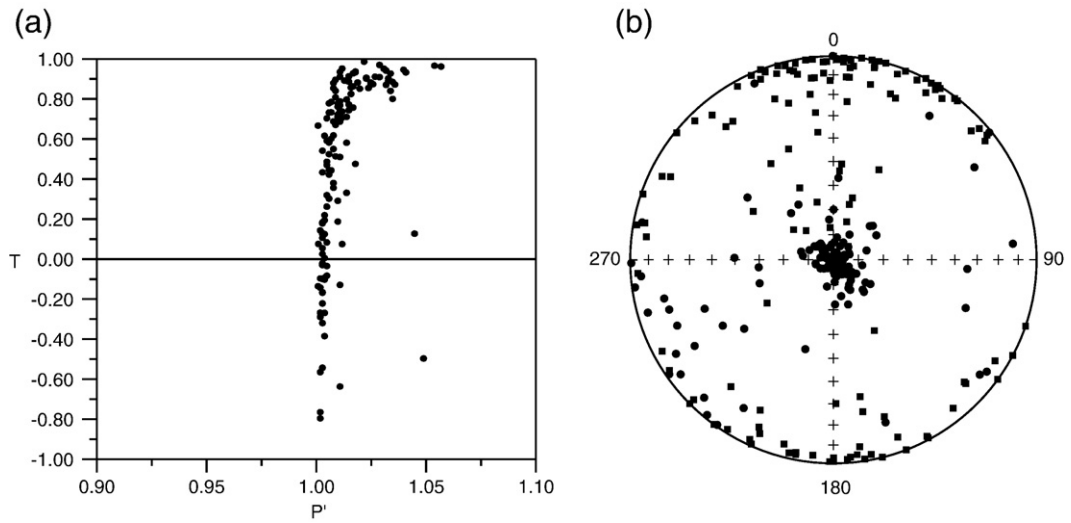


Fig. 3. (a) Anisotropy of magnetic susceptibility (AMS) plots of shape parameter *T* versus anisotropy degree *P′*, (b) stereographic projection of AMS principle axes in stratigraphic coordinates for specimens from the Yinkeng Fm. K1 (square): maximum axis, K3 (circle): minimum axis, K2 (triangle): intermediate axis.

is subdivided into three subzones according to the successive occurrence of the different morphotypes in the WP Section (Zhao et al., 2003, 2007). The *N. waageni Zone* is also subdivided into three subzones according to the successive FOs of three different types (subspecies) among the *N. waageni* group, i.e. *N. waageni* n. subsp. A (= *Neospathodus waageni eowaageni* Zhao et al., 2007 = *N. waageni* Morphotype 3 Orchard, 2007) – *N. waageni* n. subsp. B (= *N. posterolongatus* Zhao et al., 2007) – *N. waageni waageni* (= *N. waageni* Morphotype 2 Orchard, 2007). Ammonoids are common at Chaohu but the specimens from shale and mudstone in the lower part of the Lower Triassic are only preserved as molds. In comparison with the conodont *N. waageni* (the primary marker for the I/O boundary), the ammonoids show more of a facies-related distribution around the I/O boundary. More extensive biostratigraphic data on the overlying Induan are summarized in Zhao et al. (2007). The conodont specimens from the West Pingdingshan section are not clearly darkened and have a conodont alteration index of about 2.0, which is consistent with alteration index's of 1.5 to 2.0 in the general study region (Wang, 1993). The bedding in the West Pingdingshan section dips at some 78–88° towards the E, forming part of the core of the Majiashan–Pingdingshan fold.

3. Sampling and paleomagnetic procedures

A total of 347 drill-plugs were collected for palaeomagnetic investigations from the Yinkeng Formation. These were collected, using a gasoline-powered drill, and were oriented using a magnetic compass. Samples were collected at 10–15 cm intervals, covering the stratigraphic interval from just below the lithologic PTB (at 0. 33 m), to just above the base of the Olenekian (at first occurrence (FO) of *N. waageni eowaageni*, morphotype of *N. waageni*).

Fifty-two drill-plugs did not produce suitable paleomagnetic specimens for the magnetometer. The remaining 295 specimens underwent stepwise thermal demagnetization using 15–20 steps in an ASC TD-48 oven with an internal residual field less than 10 nT. Remanent magnetization was measured with a 2G cryogenic magnetometer at the palaeomagnetic laboratory of the Institute of Geomechanics, CAGS in Beijing. The magnetometer is located inside Helmholtz coils that reduce the ambient geomagnetic field to around 300 nT. Remanent component directions were determined by principal component analysis, as implemented in the Enkin suite of software. The software of Cogné (2003) and Kent et al. (1983) was

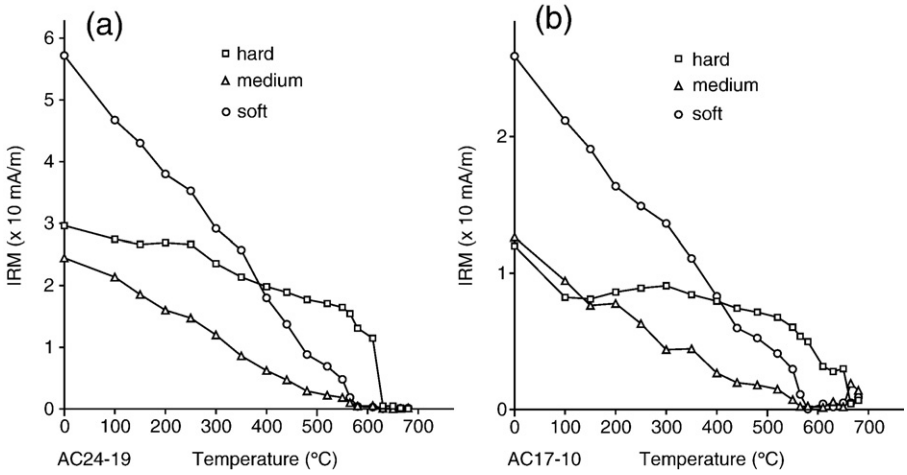


Fig. 4. Representative plots of thermal demagnetization of 3-axis isothermal remanent magnetization, using magnetizing fields of 0.12 T (soft), 0.4 T (medium) and 1.2 T (hard). These plots show Tc at ~560–580 and 630–660.

used in the analysis of the resulting demagnetization data. A KLY-4 Kappabridge susceptibility system was used to measure the anisotropy of magnetic susceptibility (AMS) of 137 specimens, prior to thermal demagnetization, to assess if these rocks have suffered substantive tectonic strain. The magnetic mineralogy of representative lithologies was studied using stepwise thermal demagnetization (Lowrie, 1990) of isothermal remanent magnetization (IRM), using IRM fields of 0.12 T, 0.4 T and 1.2 T. The IRM was imparted using an ASC scientific pulse magnetizer (Model IM-10-30).

4. Magnetic mineralogy results

The anisotropy degree (P') of the samples ranges from 1.001 to 1.057 (Fig. 3a). The mean anisotropy degree is 1.013. The anisotropy shape factor (T) varies widely, mostly independent of P', and is distributed mostly in the oblate field ($T>0$; Fig. 3a). The principal anisotropy directions (K1) are dispersed but show some evidence of a preferred N–S orientation (Fig. 3b). The K3 directions are on average perpendicular to the bedding plane. The low degree of anisotropy and the reasonably large scatter in the maximum susceptibility axis directions within the bedding plane, indicate a sedimentary origin of the fabrics (Rees and Woodall, 1975), hence these rocks have not experienced large amounts of tectonic strain, although the orientation of the mean K1 axes approximately parallel to the Majiashan–Pingdingshan fold axis is possible evidence of initial strain.

Thermal demagnetization of the soft (<0.12 T) and medium (0.12–0.4 T) fractions shows an unblocking temperature of 560–580 °C, indicative of magnetite. The hard fraction (0.4–1.2 T) shows a distinct unblocking temperature between 630 and 660 °C, which is probably indicative of hematite (Fig. 4).

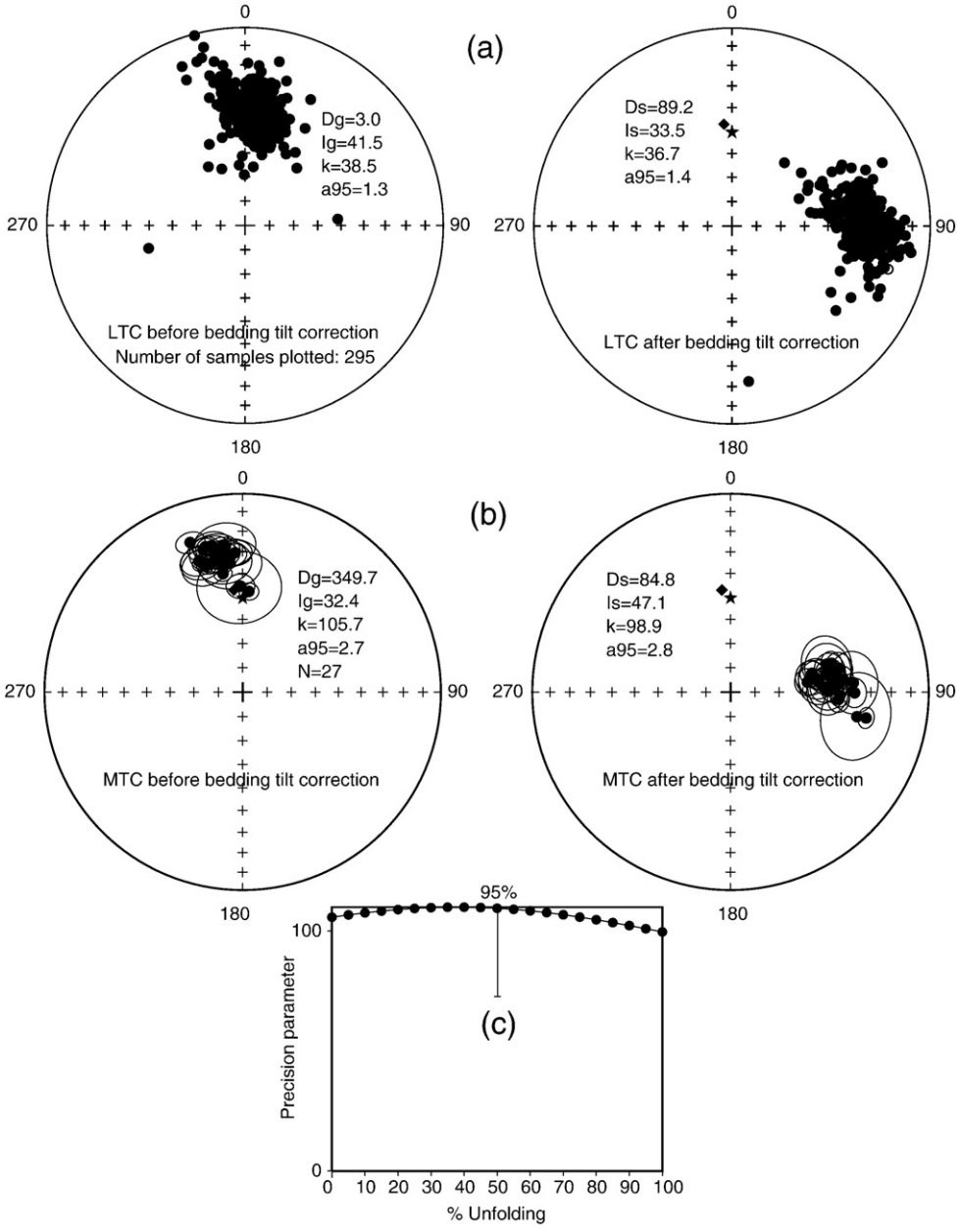


Fig. 5. (a) Equal-area projections of the low temperature (LTC or A) component. (b) The mid temperature (MTC or B) component. Star is the geocentric axial dipole field, and square the present-day field directions. (c) Maximum of the precision parameter at 40% unfolding.

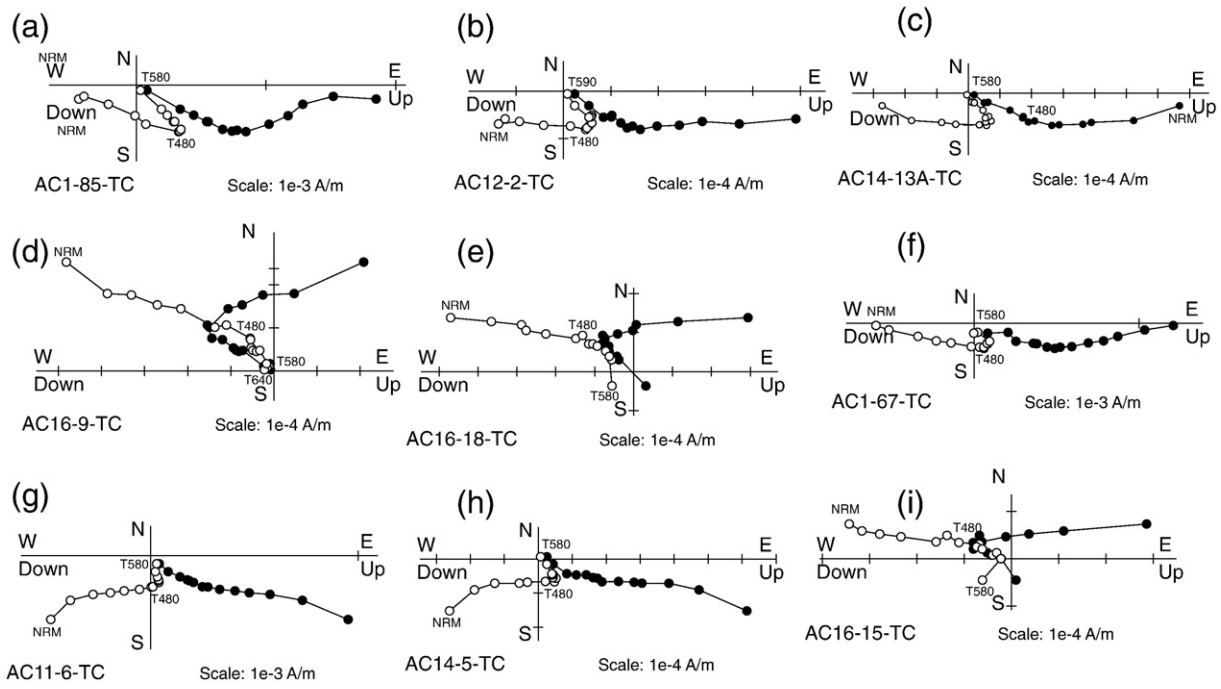


Fig. 6. Representative orthogonal demagnetization diagrams (in stratigraphic coordinates). The characteristic magnetization (component C) directions in the specimens displayed were determined by principle component line-fits. a–e: Good quality line-fit from 480 °C through the origin; f–i: poor quality line-fit from 480 °C through the origin; (d), (e), (i): normal polarity; (a)–(c), (f)–(h): reverse polarity. Demagnetization steps in °C in all plots.

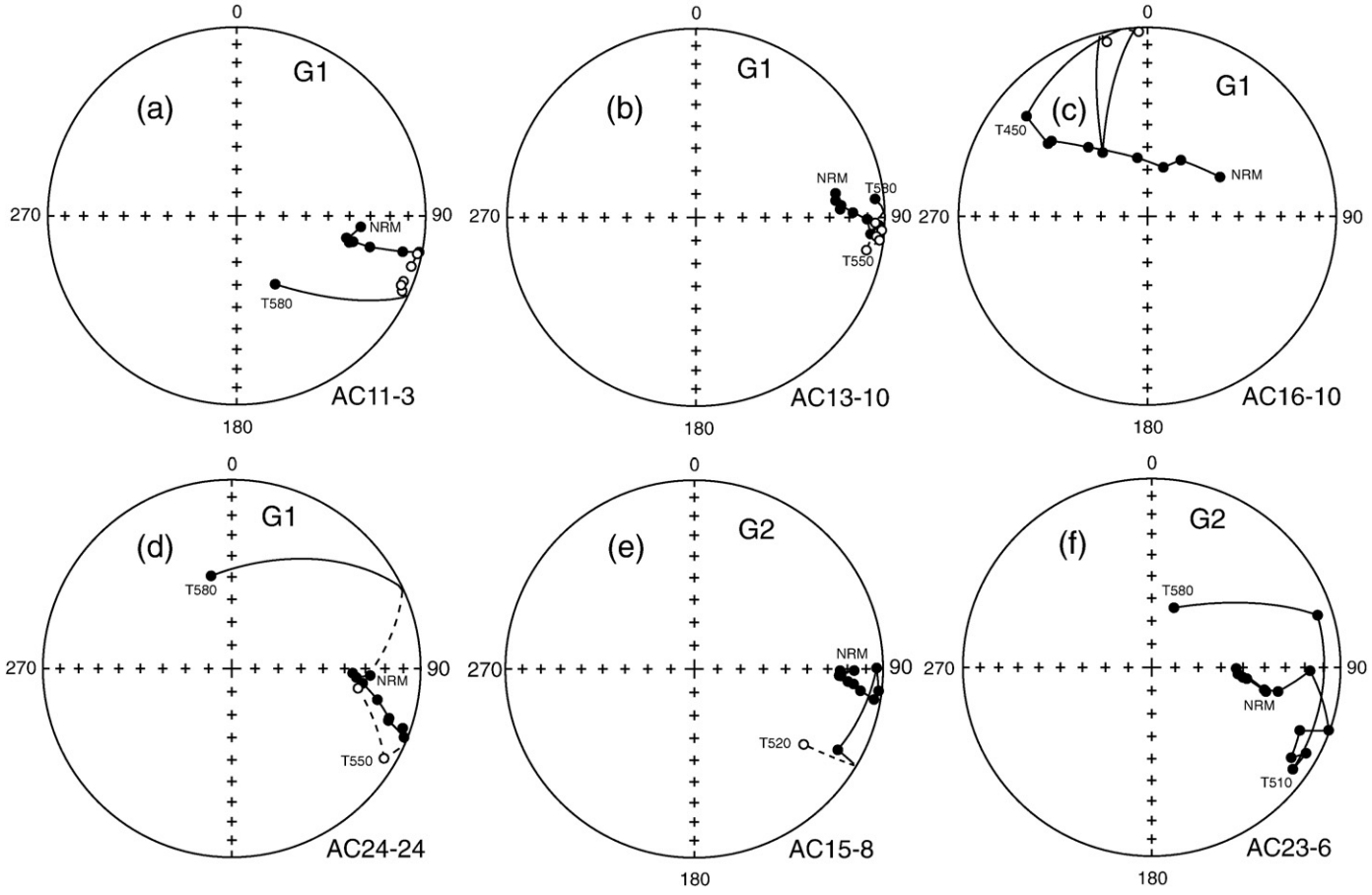


Fig. 7. Equal-area projections of the magnetization directional tracks during thermal demagnetization of representative samples with a variable content of component C (all in stratigraphic coordinates). G1: Good quality great circle data; G2: poor quality great circle (with only short paths towards directions that are consistent with component C being present). Plot c) is for a normal polarity sample, all others for interpreted reverse polarity specimens. In general, these specimens show motion along great-circle paths in a southeasterly direction, trending towards negative inclination or a northwesterly direction with a positive inclination. The great-circle path is best defined from about 480 °C to 580 °C.

5. Palaeomagnetic results

The specimens have initial natural remanent magnetization (NRM) intensities between 10^{-4} and 10^{-3} A/m. Thermal demagnetization isolated three magnetization components:

(a) Firstly a low-temperature component (component A: LTC) is isolated in all samples below about 300 °C. The mean direction of this component before tilt correction is D=3.0°, I=41.5° (N=295 and α_{95}=1.3°), and is similar to the present-day field direction (D=355.4° and I=47.3°; Figs. 5a and 6), which is inferred to be the origin of this component.

(b) A second component (component B: MTC) is determined mostly between the 300 and 480 °C demagnetization steps, by a well-defined linear segment on the orthogonal vector diagrams in nearly all specimens. This magnetization component largely dominates the NRM. This component is NNW and down-directed in geographic coordinates and easterly and down-directed in stratigraphic coordinates (Figs. 5b and 6). The Fisher precision parameter of the mean direction of the MTC component changes from 42.2 before bedding correction to 40.3 after bedding correction, indicating a slightly tighter directional dispersion in situ coordinates (Fig. 5b). However, progressive stepwise unfolding shows that the ratio of precision parameters before any correction and after progressive correction (ks/kg) passes through a maximum at a value of 40% (Fig. 5c). This characteristic may indicate that component B is probably a remagnetization acquired during the folding. The overall mean direction after 40% tilt correction is D=15.0° and I=55.2° (α_{95}=2.6, N=27 sites) and indicates a paleopole of 76.9°N, 185.4°E (α_{95}=3.1) for the synfolding magnetization. This direction has also been found by Enkin et al. (1992a) and Yang and Besse (2001) in the Permian and Triassic sediments from the Sichuan basin. They were given (at 50% unfolding) as D=17.8°, I=52.1° (α_{95}=7°, N=7 sites) and (at 60% unfolding) as D=13.0°, I=52.7° (α_{95}=3.9°, N=18 sites) respectively. The results provide a synfolding pole at 76.5°N, 174.1°E (A_{95}=3.7), which is close to our synfolding pole (76.9°N, 185.4°E, A_{95}=3.1). Its consistent presence in so remote region suggests that it constitutes a general synfolding remagnetizaton in South China Block.

(c) Thirdly a high-temperature component (component C: HTC). Only some 66% of samples showed evidence of this magnetization component, the remaining specimens were dominated by component B until complete demagnetization. Component C has a strong overlap of unblocking temperature with component B in a large majority of specimens at and above 480 °C. Only some 11% of specimens show clear linear segments, separating component C from component B on Zijderveld plots and principle component (PCA) analysis (Figs. 6 and 7). This component is mostly present between 480 and 580 °C. Its direction is of dual polarity, and is interpreted as a Triassic magnetization (Figs. 6 and 8).

Categories of demagnetization behavior were visually assigned (assisted by the PCA analysis) to several classes of demagnetization data shown by the specimens.

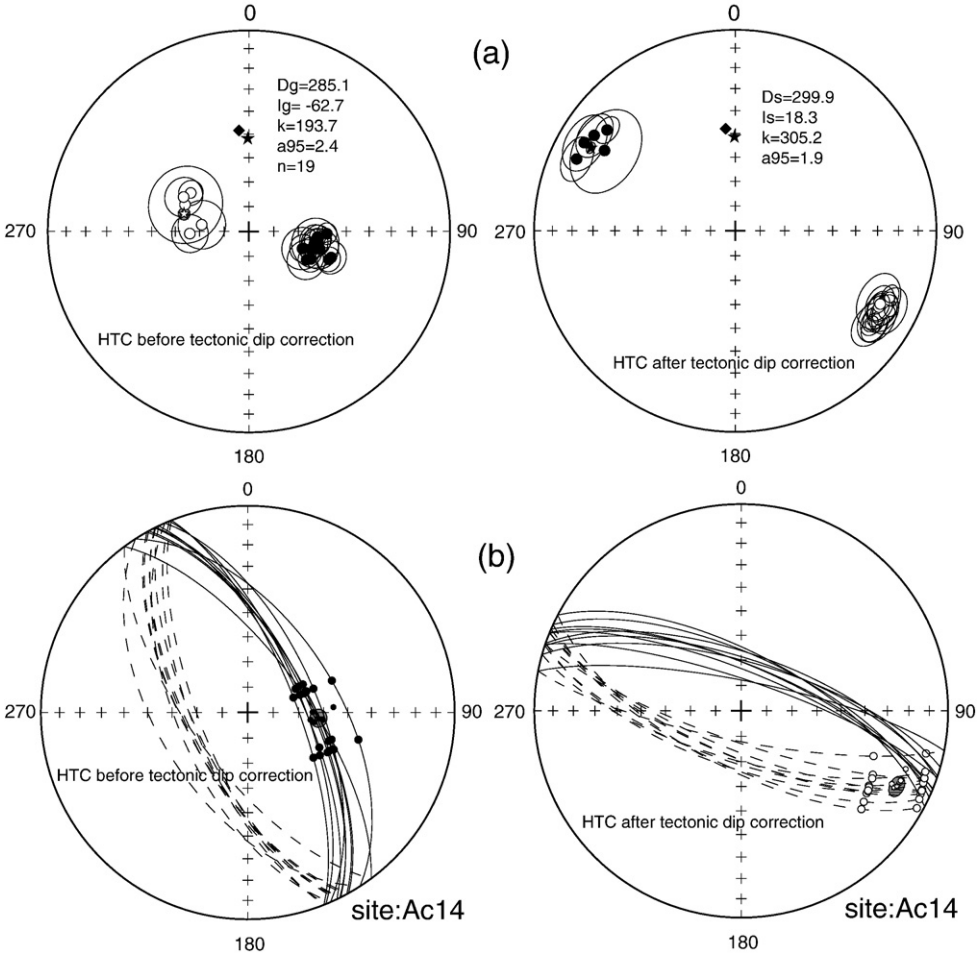


Fig. 8. (a) Equal-area stereographic projection of mean directions of high temperature component C for stratigraphic groups of specimen data. Lower (upper) hemisphere directions are marked with closed (open) symbols. (b) Equal-area stereographic projection of the site (AC14) showing the great circles (re-magnetization circles)/fixed points used to evaluate one of the means displayed in a. Ellipses are 95% confidence cones of group means. Stars = mean directions of these dual polarity magnetizations.

Firstly, a category indicating no component C could reliably be interpreted from the specimen data ('MTC only' in Fig. 9). This type of behavior is dominant in the lower 5 m of the section (which is more weathered than the overlying parts), and above 20 m in the section (Fig. 7). Some 34% of specimens possess this type of behavior.

Secondly, two classes (good and poor) of ChRM line-fit data (Figs. 6 and 9), which are exclusively present in the levels between 5 and 30 m (Fig. 9). Some 11% of samples possess this type of behavior.

Thirdly, specimen data, which showed evidence of incomplete separation of components B and C, but which showed evidence of great circle trends towards either the reverse or normal polarity directions of component C ('GC trends' in Fig. 9). Two sub-categories of good and poor behavior were evaluated, based on the amount of approach towards the component C dual polarity directions (Figs. 8 and 9). Some 55% of specimens possess this type of behavior. Great circles were predominantly fitted through the higher temperature demagnetization steps as an estimate of these great circle trends.

The reversal test (McFadden and McElhinny, 1990) has been performed on the component C mean direction. The test indicates a positive reversal test (Ra), with less than 5° between inverted antipodal mean directions (γ_{Obs}=2.3; $\gamma_{Critical}$=4.4). The tilt test on the site-mean directions of ChRM is also positive at the 95% level of

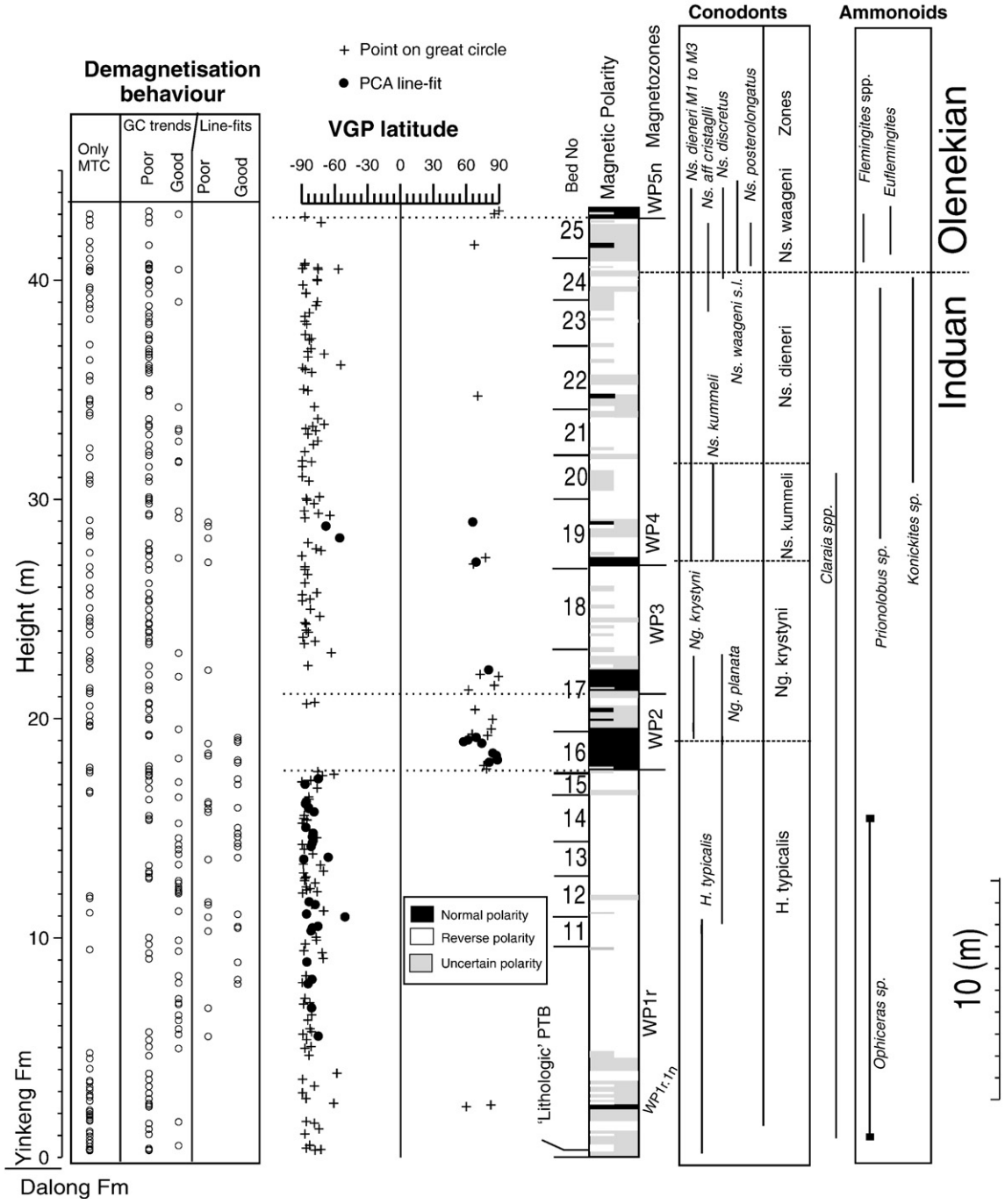


Fig. 9. Demagnetization behavior (see text), virtual geomagnetic pole (VGP) latitude, and the magnetic polarity interpreted in the West Pingdingshan section, along with summary biostratigraphic data (Zhao et al., 2007). Magnetozones defined by no adjacent specimens of the same polarity, are indicated by half bars, with full bars indicating two or more adjacent specimens of the same polarity. Full grey bars indicate adjacent specimens with only component B present. The major N–R magnetozone couplets have been labeled WP (for West Pingdingshan), for ease of description.

Table 1
Site-mean paleomagnetic results from the West Pingdingshan section at Chaohu, Anhui Province

Bed/ site	n	D_g (°)	I_g (°)	D_s (°)	I_s (°)	κ	α_{95} (°)
ac24	9	107.3	67	124.7	−17.1	30.9	5.4
ac23	11	106	63.9	122.5	−19.5	42.5	4.6
ac22	12	100.3	62.9	117.9	−19.8	23.5	5.6
ac21	8	107.8	67.2	121.8	−17.9	13.3	8.4
ac20	7	116.2	63.9	124.8	−19.2	24.9	7.9
ac19	10	303.5	−61.6	308.2	20	28.5	5
ac18	12	114	62.4	123.3	−19.2	22.2	5.2
ac17	9	267.9	−66	294.4	14.5	14.7	7.5
ac16a	7	278.1	−70.7	300.3	14.1	16.7	9.7
ac16b	7	292.8	−61.8	301.6	24.7	8.3	15.3
ac15	11	100	62.5	116.2	−21.4	10.5	9.8
ac14	14	94.6	62	115.9	−17.5	50.6	3.4
ac13	11	96.6	61.7	117.7	−16.8	25.3	5.6
ac12	10	91.7	58.2	113.3	−17.5	39.6	4.2
ac11	7	98.2	62.8	119	−16.4	28.5	6.4
ac10	10	107.6	54.7	116.8	−20.4	20.8	6.4
ac9	10	109.3	55.3	118	−20.3	30.8	4.8
ac8	10	103.9	60.3	117.5	−14.5	13.4	7.9
ac7	10	297.8	−59.9	304.1	16.4	12.3	7.8
Mean	19	285.1	−62.7	–	–	193.7	2.4
		–	–	299.9	18.3	305.2	1.9

n: number of specimens used to calculate mean; D_g, I_g, D_s, I_s: declination and inclination in geographic and stratigraphic coordinates respectively; κ: the best estimate of the Fisher precision parameter; α_{95}: the radius of the 95% cone of confidence.

confidence according to the criteria of McFadden (1990) ($Xi_2Is = 6.8$, $Xi_2Tc = 1.8$, Critical 95% = 5.07). These findings indicate that component C represents the primary Lower Triassic ChRM from the West Pingdingshan section. We fit great circles to the demagnetization data that do not possess a well defined ChRM fitted-line segments, and use these data in combination with the ChRM fitted-line segments through the Zijderveld-plot origin to determine a great-circle/fixed point mean direction using the methodology outlined by McFadden and McElhinney (1988), as implemented in the Cogné (2003) software (Fig. 8). The averaged site-mean direction for component C is $D_g = 285.1°$, $I_g = -62.7°$, $\kappa_g = 193.7$, $\alpha_{95} = 2.4°$ before tilt correction, and $D_s = 299.9°$, $I_s = 18.3°$, $\kappa_s = 305.2$, $\alpha_{95} = 1.9°$, $N = 19$ after tilt correction (Fig. 8a, Table 1). The average direction of each group of sites was calculated using the great circles (re-magnetization circles)/fixed points to evaluate one of the means displayed in Fig. 8a, for example the samples in bed AC14 (Fig. 8b). The determined paleopole lies at 30.3°N, 19.9°E with $A_{95} = 1.4°$ (dp/dm = 1.0/2.0). The paleolatitude of 9.4° for the section is not significantly different (at 95% confidence level) with the 11–12° predicted for the stable South China Block (Heller et al., 1988, 1995; Steiner et al., 1989). Our early Triassic pole study comes from different lithologies (mudstone and limestone) with a magnetization carried by magnetite and passes successfully both the fold test and reversal test. These findings indicate the component C may likely represent the primary nature of the ChRM from the West Pingdingshan section.

To constrain the paleogeographic affinities of the studied area, paleopoles for the synfolding B component and the characteristic C component are compared with the Mesozoic APW path of the South China Block (SCB) (Fig. 10). The paleopole calculated from the 40% untilted B-component mean direction, is situated in the confidence ellipse of the late Cretaceous pole from the SCB (Fig. 10), which may correspond to the late phase of the Yanshanian orogeny. Yang and Besse (2001) proposed that the so called folding-associated remagnetization was probably acquired between Middle Jurassic and late Cretaceous. However, the C component paleopole shows a large counter-clockwise rotation (101.2° ± 3.3°) with respect to early Triassic/late Permian poles from the stable SCB (Fig. 10). Similar anticlockwise rotations occur further south adjacent to the Tanlu fault (Tan et al., 2007). These are probably associated with local rotation of the Chaohu area, during docking with the nearby North China block, along the Tanlu fault. Previous paleomagnetic studies have already addressed the collision history of the North China block (NCB) and SCB (Enkin et al., 1992b; Gilder and Courtillot, 1997). These concluded that the two blocks were welded together during the early to mid Jurassic, whereas Yang and Besse (2001) proposed that both blocks were already sutured by the mid Jurassic.

6. Magnetostratigraphy

The ChRM directions were converted to VGP latitude (displayed as filled circle in Fig. 9). For those specimens possessing great-circle behavior, the point on the fitted great circle closest to the section mean direction was used to calculate the VGP latitude (displayed as cross in Fig. 9). The magnetic polarity of the section is dominated by reverse polarity, with three normal polarity magnetozones represented by three or more specimens (WP2n, WP3n and WP4n). Normal magnetozones WP1r.1n and WP5n are defined by only two specimens. A number of tentative normal polarity sub-magnetozones, represented by a single specimen are also present within WP4r, with only that in bed 19 defined by PCA line-fit data (Fig. 9).

The magnetic polarity at the correlated base of the Olenekian (i.e. the FO of morphotype *N. waageni eowaageni* in subbed 24–16) is probably reverse, since underlying sub-bed 24–15 and overlying sub-bed 24–17 are reversed, with both sampled intervening beds not containing any evidence of Triassic component C. The base of the overlying normal magnetozone W5n is ~2.5 m above the FA of *N. waageni eowaageni*, although a substantial number of specimens in the base of bed 25 possess no component C. Hence, the base of magnetozone WP5n provides a secondary marker a little above the base of the Olenekian in the West Pingdingshan section.

The dominance of reverse polarity in the lower part of the section (i.e. that below bed 15, Figs. 9 and 11) would suggest, according to the magnetostratigraphy, that this interval is entirely late Griesbachian, since no substantive evidence of the equivalent of normal polarity magnetozone LT1n (using chron nomenclature of Hounslow et al., 2008), which characterizes the basal Induan, occurs in the base of the

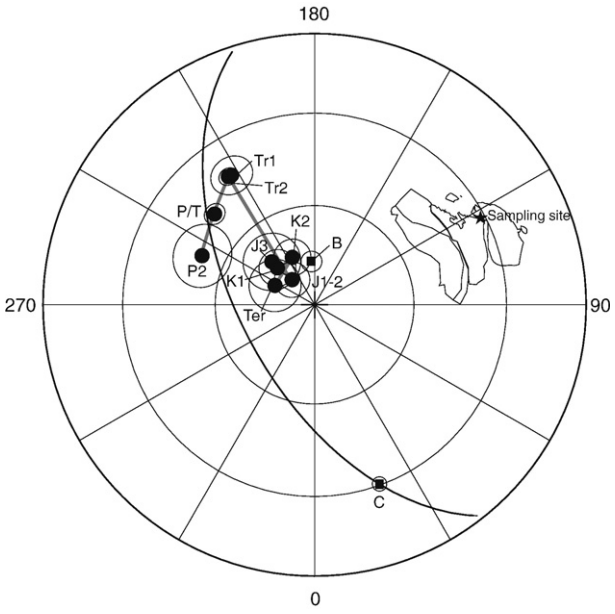


Fig. 10. The Mesozoic APW path of the South China Block in stereographic projection from compilation of Yang and Besse (2001), compared with the poles calculated from the B and C components isolated at the Chaohu section. The small circle through the C pole is the paleocolatitude calculated from the C component paleolatitude. Age symbols: P2, late Permian; P/T, Permian–Triassic boundary; T1, Lower Triassic; T2, Middle Triassic; J1–2, Lower–Middle Jurassic; J3, Upper Jurassic; K1, Lower Cretaceous; K2, Upper Cretaceous; Ter, Tertiary.

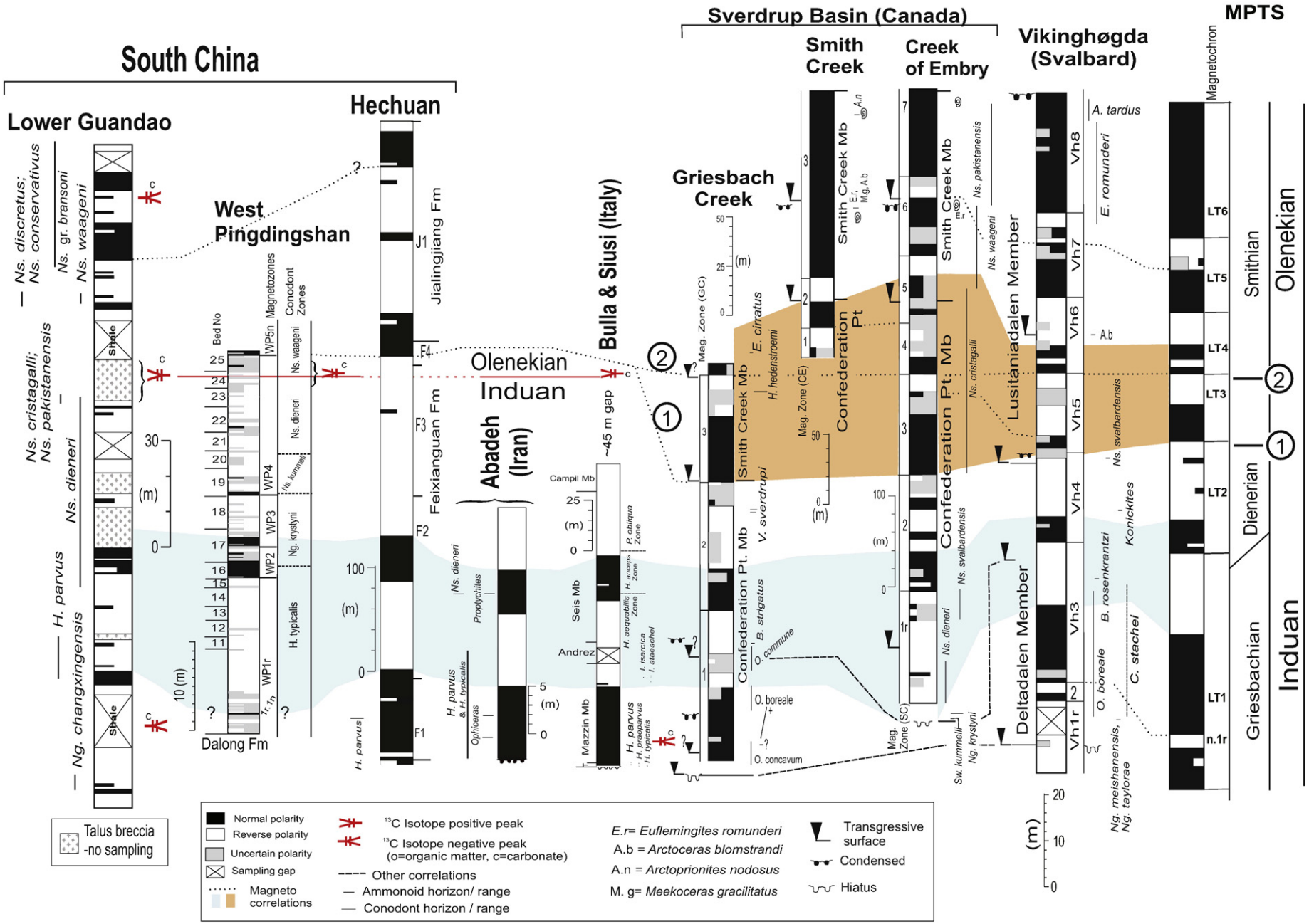


Fig. 11. Comparison of the magnetostratigraphy at Chaohu with other studies of marine sections through the latest Permian and Lower Triassic. Data for polarity columns: Guandao (Lehrmann et al., 2006), Hechuan (Steiner et al., 1989), Abadeh (Gallet et al., 2000), and Bulla/Siusi (Scholger et al., 2000; Perri and Farabegoli, 2003; Horacek et al., 2007). Griesbach Creek, Smith Creek and Creek of Embry (Ogg and Steiner, 1991 modified by Hounslow et al., 2008; Beatty et al., 2008; Orchard, 2008; Baud et al., 2008), Vikinghøgda and composite magnetic polarity timescale (MPTS) from Hounslow et al. (2008).

section (Fig. 11). It is perhaps possible that the normal magnetozone WP1r.1n, some 2.0 m above the "PTB set" (Peng et al., 2001), may represent part of LT1n and the remainder of the basal Griesbachian normal magnetozone is obscured by the scarcity of polarity data in this interval (Fig. 9). The basal part of the West Pingdingshan section has not had much detailed biostratigraphic work completed on it, but neighboring sections such as the North Pingdingshan and West Majiashan sections have been studied in detail at the boundary, with no apparent breaks in sedimentation.

7. Global magnetostratigraphic correlations

The magnetostratigraphy of the Gaundao section (in Guanxgi province, S. China) is ambiguous around the Induan–Olenekian boundary (IOB), but there the positive carbon isotopic excursion allows approximate correlation between the Gaundao, West Pingdingshan and Bulla/Suisi sections (Payne et al., 2004; Horacek et al., 2007; Tong et al., 2007; Richoz et al., 2007; Fig. 11). The magnetostratigraphy of the upper part of the West Pingdingshan section, has a close correspondence in polarity style to that at Hechuan in S. China, as does the reverse-polarity dominated interval at Guandao, which covers the upper range of the conodont *N. dieneri* (Fig. 11). The most complete magnetostratigraphy across the IOB occurs in the Boreal realm from the Sverdrup Basin in Canada and Spitsbergen in arctic Norway (Ogg and Steiner, 1991; Hounslow et al., 2008; Fig. 11). However, correlation of the IOB onto the magnetostratigraphy from the boreal sections is problematic with two possible solutions.

The oldest occurrence of *N. waageni* with respect to the boreal ammonoid zones, is some distance above the FO of *H. hedenstroemi* in Siberian sections (Dagis, 1984), whereas more widespread occurrences of *N. waageni* first occurs within the overlying *Euflemingites romunderi* Zone in Canadian sections (Orchard and Tozer, 1997; Orchard, 2008). This suggests that the IOB probably occurs within the lower part of the *H. hedenstroemi* Zone. In Canadian sections, this relationship cannot be confirmed, since conodonts have not been recovered from beds bearing *H. hedenstroemi* (Orchard, 2008). The conodont *N. cristagalli*, whilst having a restricted range in Chaohu near the IOB has a range in Canadian sections from the late Griesbachian Strigatus Zone into the late Dienerian Sverdrupi Zone (Orchard, 2008). Likewise, *N. pakistanensis*, whilst commonly associated with Romunderi Zone fauna in Canada, is known from Spitsbergen sections associated with, *V. spitzbergensis* (proxy for the Sverdrupi Zone; Nakrem et al., 2008). Similarly, the three morphotypes of *N. dieneri*, described from the late Dienerian to early Smithian strata at Chaohu are known from Canadian sections to range through the Dienerian Candidus and Sverdrupi Zones (Orchard, 2008). However, this does suggest that the reverse polarity interval covering beds 18–24 at West Pingdingshan (Fig. 11) probably represents the equivalent of magnetozone LT2r (GC2r at Griesbach Creek; Sc2r at Creek of Embry and Vh4r at Vikinghøgda; Fig. 11) in the Boreal composite magnetic polarity timescale (MPTS). Hence, the magnetozone WP5n at West Pingdingshan is probably the equivalent of magnetozone LT3n in the boreal MPTS (option ① in Fig. 11), and hence the first possibility is that the IOB is within the uppermost part of LT2r. Two features support this possibility; firstly, this correlation is consistent with the magnetostratigraphy, in that no substantive normal magnetozone is detected in beds 18 to 24 at West Pingdingshan. Secondly, at both the Griesbach Creek and Vikinghøgda sections, close to the lower boundary of GC3n and Vh5n (LT3n in MPTS) is a major transgressive surface, which may be the equivalent of that seen at Chaohu and Mud close to the IOB (Guo et al., 2007; Krystyn et al., 2007).

A second, higher placement of the IOB, relative to the magnetostratigraphy is also possible (option ② in Fig. 11). Since the ammonoid control at Griesbach Creek is based on spot occurrences, it is not clear where the base of the *H. hedenstroemi* Zone is located, it may be that it is located in the upper part of the Smith Creek Member in the upper part of magnetozone GC3 (Fig. 11). Consequently, magnetozone WP5n may be the equivalent of LT4n in the MPTS (Fig. 11). This option is more compatible with conodont to ammonoid correlations, which suggest at Creek of Embry that the Dienerian extends to high in the Confederation Point Member. However, a consequence of this biostratigraphic-driven correlation is that the equivalent of LT3n in the MPTS appears to be unconvincingly detected at West Pingdingshan — possibly represented by WP4n in bed 19? (Fig. 11).

With these uncertainties in correlation of marine sections in mind, its premature to attempt mapping of the cyclo-magnetostratigraphy from the Buntsandstein (Szurlies, 2007) into the marine sections. In addition much reliance has been placed on relating the Italian sections at Bulla and Siusi (Scholger et al., 2000) to age-calibrate the Buntsandstein cycles and lithostratigraphy (Szurlies et al., 2003; Szurlies, 2007), yet the isotopic curves from these Italian sections clearly locate the IOB higher in the Bulla/Siusi sections than that used by Szurlies (2007).

8. Conclusion

A dual polarity Triassic magnetization can be extracted from the West Pingdingshan section, in spite of minor associated tectonic deformation, and partial remagnetization. The early Triassic magnetization is masked by a strong Jurassic–Cretaceous overprint magnetization, acquired post-tilting, which variably masks the early Triassic magnetization. Nevertheless some 66% of specimens display evidence in the demagnetization diagrams of characteristic polarity, either through conventional magnetization component isolation, or great circle trends. The correlated conodont marker for the base of the IOB in the GSSP at Mud (FA of *N. waageni s. l.*) indicates that at West Pingdingshan the IOB is some 2.5 m below a normal magnetozone, which provides a secondary marker, for mapping the base of the Olenekian into successions without conodonts. Around the IOB the West Pingdingshan section can be correlated confidently to other Tethyan Lower Triassic successions, but which lack the corroborative magnetostratigraphic details shown at Chaohu. Problems remain near the Permian–Triassic boundary in the West Pingdingshan section, due to inadequate recovery of Triassic magnetizations from specimens. Problems also remain in relating the Induan–Olenekian boundary interval into the boreal sections in Canada and Spitsbergen with magnetostratigraphy.

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